AYE
TECH HUB

• MEP ENGINEERING • LESSON 02

MEP in Building Construction

Coordination · Zones · Risers · Clash Detection · Drawings

Lesson 02 of 30 | MEP Engineering Program

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- 7-stage MEP coordination workflow from design to construction
- The Rule of 10: clash found in design costs 100× less than on site
- MEP spatial zones — ceiling void, riser shaft, plant room, roof
- The MEP drawing set — GA plans, schematics, details, as-builts
- MEP specification structure and BIM Execution Plan (BEP)

How MEP Fits into Building Construction

MEP design does not happen after the building is designed — it happens **in parallel with architecture and structure**. The three disciplines share the same physical space inside a building and must be coordinated to avoid clashes, ensure maintainability, and meet programme dates. On a typical commercial project, MEP installation represents **35–50% of the total construction programme**.

The single biggest source of MEP rework and cost overrun is **coordination failure** — a duct clashing with a structural beam, a pipe running through a column, or insufficient ceiling void depth. Modern projects solve this with Building Information Modelling (BIM), but understanding the underlying coordination principles is essential regardless of the tools used.

The MEP Coordination Workflow

Figure 1 — MEP Coordination Workflow (Design to Construction)

Stage	Who Leads	Key Output	Common Issues
1 — Arch Drawings	Architect	Floor plans, sections, reflected ceiling plans	Insufficient ceiling void; no MEP zones indicated
2 — Structural Model	Structural Engineer	3D structural BIM model; beam depths; slab penetrations	Beams clash with duct routes; no penetration sleeves
3 — MEP Design	M, E, P consultants	Individual discipline 3D models (Revit MEP or similar)	Over-sized routes; no allowance for insulation thickness
4 — Federated BIM	MEP BIM Coordinator	Single model combining Arch + Structure + MEP	Model origins not aligned; LOD inconsistent
5 — Clash Detection	MEP Coordinator (Navisworks)	Clash report with location, type, and responsible party	100s of clashes on first run — prioritise hard clashes first
6 — Resolved Model	All disciplines jointly	Updated models with clashes resolved and signed off	Re-routing may change equipment sizes or programme
7 — Construction Issue	Lead Consultant	IFC / PDF drawings issued for construction and fabrication	Late changes after issue cause expensive re-work on site

Rule of 10: A clash found in design costs 1× to fix. Found in BIM coordination costs 10×. Found on site costs 100×. Every hour spent on BIM coordination saves days of rework.

MEP Spatial Zones in Buildings

One of the first tasks in MEP design is claiming space. MEP systems need **dedicated zones** that architects and structural engineers must respect from the very start. These zones are agreed in the building services strategy at feasibility stage and protected throughout design development.

Figure 2 — MEP Spatial Zones in a Multi-Storey Building

MEP Zone Requirements by Building Type

Zone	Typical Depth / Size	What Goes There	Coordination Risk
Ceiling Void (Typical Floor)	500–900 mm (office) 900–1 400 mm (hospital)	Supply/return ductwork, chilled water pipes, electrical cable trays, sprinkler mains, data cabling	Duct and pipe crossing conflicts; structural beam intrusion; access for maintenance
Riser Shaft	1.5–6 m² per shaft (depends on building height)	Vertical pipe stacks (HWS/CWS), electrical bus-risers, data/comms trunking, fire mains	Shaft size underestimated at concept; fire compartmentation breaks
Basement Plant Room	3–8% of gross floor area (typical commercial)	Chillers, boilers, cooling towers, main LV switchgear, water tanks, pumps, generators	Insufficient height for equipment; no clear maintenance access routes
Roof Plant Area	10–20% of roof area	AHUs, cooling towers, condensers, solar PV, satellite dishes, smoke extract fans	Structural loading; acoustic impact on top-floor occupants; access safety
Service Corridor	900–1 200 mm clear walkway alongside plant	Access route for maintenance engineers; may contain small distribution boards	Often sacrificed to gain lettable area — creates major maintenance problems

> ***Riser shaft tip: Always design riser shafts 20–30% larger than the calculated minimum. Buildings change use over their 50-year life — future tenants will need to add services, and retrofitting a new riser shaft through occupied floors costs 10× the price of the extra shaft space at construction.***

The MEP Drawing Set and Specifications

The MEP drawing set is the primary communication tool between the design team, the contractor, and the building owner. Each drawing type has a specific purpose and audience. A complete MEP drawing set for a commercial project can run to hundreds of sheets across all three disciplines.

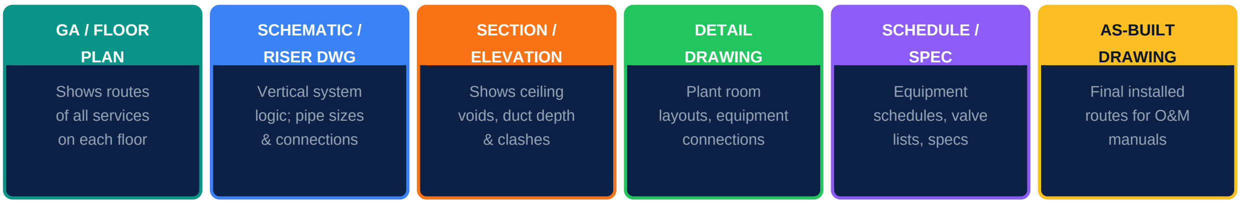

Figure 3 — The MEP Drawing Set

MEP Drawing Standards and Notation

Drawing Element	Standard	Mechanical	Electrical	Plumbing
Line weight convention	BS/ISO 128	Heavy = mains; light = branches	Heavy = power; dashed = data	Heavy = soil; dashed = vent
Service identification colour	BS 1710 / ANSI A13.1	Green = chilled water; orange = heating; grey = ductwork	Yellow = power; blue = data; red = fire alarm	Blue = cold water; red = hot water; black = drainage
Pipe/duct size notation	ISO 6708	DN150 (pipe diameter); 600×400 (duct W×H)	Cable ref in schedule (e.g. 4×16 mm^2 Cu)	DN100 (soil); 22 mm (copper CWS)
Flow direction	All disciplines	Arrow on pipe/duct centreline	Arrow on cable route	Arrow on pipe centreline
Equipment tag format	Project-specific	e.g. AHU-01-B1 (AHU, floor 01, unit B1)	e.g. DB-L3-02 (distribution board, level 3)	e.g. PRV-03-05 (pressure reducing valve)

MEP Specification Structure

Spec Section	Covers
General Requirements	Standards to comply with (ASHRAE, IEC, CIBSE), design intent, responsibilities
Mechanical — Section M	HVAC equipment performance, ductwork class, pipe class, insulation spec, testing
Electrical — Section E	Cable types, containment, luminaire performance, switchgear ratings, earthing
Plumbing — Section P	Pipe materials (copper/CPVC/stainless), jointing, water treatment, testing pressure
Commissioning — Section C	TAB (Test, Adjust, Balance) procedure, witnessing, O&M; manual content requirements
BIM Execution Plan (BEP)	Model LOD per stage, file naming, coordination meetings, clash tolerance (0 hard clashes)

What comes next — MEP-003: Reading MEP Drawings and Specifications — how to navigate a real MEP drawing set, understand symbols, read schematics, and extract the information you need for design, procurement, or site work.